Volatility Forecasting with Double Markov Switching GARCH Models

CATHY W. S. CHEN[1]*, MIKE K. P. SO[2] AND EDWARD M. H. LIN[1]

[1]*Feng Chia University, Taiwan*
[2]*Hong Kong University of Science and Technology, Hong Kong*

ABSTRACT

This paper investigates inference and volatility forecasting using a Markov switching heteroscedastic model with a fat-tailed error distribution to analyze asymmetric effects on both the conditional mean and conditional volatility of financial time series. The motivation for extending the Markov switching GARCH model, previously developed to capture mean asymmetry, is that the switching variable, assumed to be a first-order Markov process, is unobserved. The proposed model extends this work to incorporate Markov switching in the mean and variance simultaneously. Parameter estimation and inference are performed in a Bayesian framework via a Markov chain Monte Carlo scheme. We compare competing models using Bayesian forecasting in a comparative value-at-risk study. The proposed methods are illustrated using both simulations and eight international stock market return series. The results generally favor the proposed double Markov switching GARCH model with an exogenous variable.

KEY WORDS heteroscedastic models; Markov chain Monte Carlo; regime-switching models; value at risk; volatility.

*Corresponding author. Address: Department of Statistics, Feng Chia University, Taichung, Taiwan. Email: chenws@fcu.edu.tw; Tel: 886 4 2451 7250 ext. 4412; Fax: 886 4 2451 7092.

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INTRODUCTION

Nonlinearity in financial time series has often been detected by various tests for asymmetry or threshold nonlinearity in the literature. Two main classes of regime-switching models have been proposed to model nonlinear behavior (see Franses and van Dijk, 2000). The most important member of the first class of models, which assumes that the regime at time t can be determined by an observable variable, is the threshold autoregressive (TAR) model of Tong (1978); see also Tong and Lim (1980) and Tong (1983, 1990). The second class of regime-switching models assumes that the dynamic properties depend on the present regime, with the regimes being realizations of a hidden Markov chain with finite state space. Markov switching models were introduced by Hamilton (1988, 1989) and continue to gain popularity in financial time series analysis.

In practice, many financial time series exhibit asymmetry not only in the mean but also often in the variance. The first class of regime-switching volatility models assumes that the regime can be determined by an observable variable, such as the exponential GARCH and threshold GARCH models of Nelson (1991) and Zakoian (1994), respectively. The GJR-ARCH model, a nonlinear threshold model first used by Glosten, Jaganathan and Runkle (1993), employs a dummy variable for negative shocks in the volatility equation. The second class of regime-switching models for volatility assumes that the regime is determined by an unobserved variable. A generalization to the Markov switching GARCH model was developed by Gray (1996) and subsequently extended by Klaassen (2002). Various versions were applied in Kim (1993), Cai (1994), Hamilton and Susmel (1994), Dueker (1997), and Yoo (2004). A Markov switching stochastic volatility model was also studied in So, Lam and Li (1998). For details see Franses and van Dijk (2000). Many of these studies use an ARCH specification instead of a GARCH to avoid the problem of path dependence of the conditional variance, which renders computation of the likelihood function infeasible.

Li and Li (1996) appear to be the first to attempt to model mean and volatility asymmetry together. They propose a double threshold ARCH model for financial market returns. Brooks (2001) extends this model to a double threshold GARCH (DT-GARCH) model. However, both the models in Li and Li (1996) and in Brooks (2001) suffer from the weakness that the threshold variable is subjectively fixed. Chen and So (2006) introduce a threshold heteroscedastic model with exogenous variables where the threshold variable is defined by a weighted average of auxiliary variables. Additionally, they assume that the error term follows a standardized t distribution with mean 0 and variance 1. The papers mentioned here all focus on regimes determined by observable variables. In contrast, this paper models mean and volatility asymmetry together using a double Markov switching GARCH (DMS-GARCH) model for financial time series. We propose a Bayesian framework, employing Markov chain Monte Carlo (MCMC) computational techniques and the Metropolis-Hastings (MH) method (Metropolis *et al.*, 1953; Hastings, 1970) for inference and model comparison based on value-at-risk forecasting performance.

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Tsay (2005) proposes a simple two-state Markov switching model with different risk premiums and different GARCH dynamics for an asset return. Considerable empirical evidence in the literature shows that the error distribution is fat-tailed. Therefore, assuming normality of the error term, as in Tsay (2005), may not be appropriate. A key feature of the proposed DMS-GARCH model is that it allows for some (or all) of the parameters to switch across different states according to a Markov process governed by state variables. We use a hidden Markov chain to determine the switching from one state to another in the conditional mean and variance equations.

The DMS-GARCH model is defined as follows:

$$y_t = \begin{cases} \phi_0^{(1)} + \sum_{i=1}^{p} \phi_i^{(1)} y_{t-i} + \sum_{j=1}^{q} \psi_i^{(1)} x_{t-j} + a_t & \text{if } s_t = 1, \\ \phi_0^{(2)} + \sum_{i=1}^{p} \phi_i^{(2)} y_{t-i} + \sum_{j=1}^{q} \psi_i^{(2)} x_{t-j} + a_t & \text{if } s_t = 2, \end{cases}$$

$$a_t = \sqrt{h_t}\epsilon_t, \qquad \epsilon_t \overset{\text{i.i.d.}}{\sim} D(0,1), \tag{1}$$

$$h_t = \begin{cases} \alpha_0^{(1)} + \sum_{i=1}^{g} \alpha_i^{(1)} a_{t-i}^2 + \sum_{i=1}^{k} \beta_i^{(1)} h_{t-i} & \text{if } s_t = 1, \\ \alpha_0^{(2)} + \sum_{i=1}^{g} \alpha_i^{(2)} a_{t-i}^2 + \sum_{i=1}^{k} \beta_i^{(2)} h_{t-i} & \text{if } s_t = 2, \end{cases}$$

where the error terms ε_t are independently distributed from an unspecified distribution D with mean 0 and variance 1, y_t is the return data, h_t is the conditional volatility of $y_t \mid y_1, \ldots, y_{t-1}$, and x_t is an exogenous variable. The sequence $\{s_t\}$ is assumed to be a stationary, irreducible Markov process with discrete state space $\{1, 2\}$ and transition probabilities:

$$\Pr\left(s_t = j \mid s_{t-1} = i\right) = p_{ij},$$

where $p_{11} + p_{12} = 1$, $p_{21} + p_{22} = 1$, and $0 < p_{ij} < 1$. Note that a small p_{ij} indicates that the series has a tendency to stay in the ith regime, and the expected sojourn time is $1/p_{ij}$. To allow dynamic volatility, we use a GARCH formulation in the equation for the conditional variance h_t. Standard restrictions on the variance parameters apply:

$$\alpha_0^{(j)} > 0, \quad \alpha_i^{(j)}, \beta_i^{(j)} \geq 0, \quad \text{and } 0 < \sum_{i=1}^{g} \alpha_i^{(j)} + \sum_{i=1}^{k} \beta_i^{(j)} < 1 \quad \text{for} \quad j = 1, 2. \tag{2}$$

These constraints ensure that the series $\boldsymbol{y} = (y_1, \ldots, y_n)'$ is stationary and that each volatility h_t is positive. We adopt a Bayesian approach which allows us to estimate the mean parameters, variance parameters, transition probabilities, and states simultaneously. Specifically, we generate approximated samples from the posterior distribution of unknown parameters and states via MCMC methods.

The major difference between the DMS-GARCH in (1) and a two-regime DT-GARCH model is that the latter defines the regimes by a threshold variable. For example, Brooks (2001) assumes the threshold variable to depend on past realizations, whereas our DMS-GARCH model uses a stochastic process to define the unknown states. In practice, the stochastic nature of the states implies that one is never certain about the state that y_t and h_t are in at any given time in the DMS-GARCH model.

BAYESIAN INFERENCE

Under the DMS-GARCH model, we define the data vector $\boldsymbol{y}^{r+1,n} = (y_{r+1}, \ldots, y_n)'$ and the state vector $\boldsymbol{S}^{r+1,n} = (s_{r+1}, \ldots, s_n)'$. Suppose $r = \max\{p, q, g, k\}$ and define the mean parameter vector $\boldsymbol{\phi}_j = (\phi_0^{(j)}, \ldots, \phi_p^{(j)}, \psi_1^{(j)}, \ldots, \psi_q^{(j)})'$ and the variance parameter vector $\boldsymbol{\alpha}_j = (\alpha_0^{(j)}, \ldots, \alpha_g^{(j)}, \beta_1^{(j)}, \ldots, \beta_k^{(j)})'$ for $j = 1, 2$. Also, let $\boldsymbol{\Theta}$ denote the unknown parameter vector $(\boldsymbol{\phi}_1', \boldsymbol{\phi}_2', \boldsymbol{\alpha}_1', \boldsymbol{\alpha}_2', p_{11}, p_{22})'$. The likelihood function of our model is:

$$p\left(\boldsymbol{y}^{r+1,n} \mid \boldsymbol{S}^{r+1,n}, \boldsymbol{\Theta}\right) = \prod_{t=r+1}^{n} \left[\frac{1}{\sqrt{h_t}} p_\varepsilon\left(\frac{y_t - \mu_t}{\sqrt{h_t}}\right)\right], \tag{3}$$

where

$$\begin{aligned} \mu_t &= \phi_0^{(j)} + \sum_{i=1}^{p} \phi_i^{(j)} y_{t-i} + \sum_{j=1}^{q} \psi_i^{(j)} x_{t-j}, \\ h_t &= \alpha_0^{(j)} + \sum_{i=1}^{g} \alpha_i^{(j)} a_{t-i}^2 + \sum_{i=1}^{k} \beta_i^{(j)} h_{t-i} \quad \text{for} \quad j = 1, 2, \end{aligned}$$

and p_ε denotes the probability density function of ε_t.

To account for the leptokurtosis observed in financial time series, we propose a heavy-tailed distributed error term. Assuming ε_t follows $D = t_\nu^*$ (i.e., the standardized t distribution with mean 0, variance 1, and degrees of freedom ν), the likelihood function of the DMS-GARCH model becomes:

$$p\left(\boldsymbol{y}^{r+1,n} \mid \boldsymbol{S}^{r+1,n}, \boldsymbol{\Theta}\right) = \prod_{t=r+1}^{n} \left\{ \frac{\Gamma(\frac{\nu+1}{2})}{\Gamma(\frac{\nu}{2})\sqrt{(\nu-2)\pi}} \frac{1}{\sqrt{h_t}} \left[1 + \frac{(y_t - \mu_t)^2}{(\nu-2)h_t}\right]^{-\frac{\nu+1}{2}} \right\}. \tag{4}$$

Bayesian inference requires setting prior distributions for the unknown parameters. We assume that $\boldsymbol{\phi}_j$, $j = 1, 2$, follows a multivariate normal distribution, $N(\boldsymbol{\mu}_{\phi_j}, \boldsymbol{\Sigma}_{\phi_j})$, where $\boldsymbol{\mu}_{\phi_j}$ is the mean vector and $\boldsymbol{\Sigma}_{\phi_j}$ is the variance-covariance matrix. We adopt priors that are mostly uninformative over the restricted parameter region in (2). The prior for $\boldsymbol{\alpha}_j$ is flat over (2):

$$p(\boldsymbol{\alpha}_j) \propto I(0 < \alpha_0^{(j)} < b) I(A_j).$$

where b is a positive integer specified by the user and A_j satisfies the restrictions in (2). Furthermore, we employ a conjugate prior distribution, Beta(u_{i1}, u_{i2}), for p_{ii} given by:

$$p\left(p_{ii}\right) \propto p_{ii}^{u_{i1}-1}(1 - p_{ii})^{u_{i2}-1}, \; i = 1, 2. \tag{5}$$

Multiplying the likelihood in (4) and the priors, the Bayes rule leads to suitable conditional posteriors for MCMC sampling. The posteriors are in general not of a standard

form and require us to employ techniques, such as the MH method to achieve the desired sample. We use the following parameter groupings: the mean parameters in each state $\boldsymbol{\phi}_j$; the variance parameters in each state $\boldsymbol{\alpha}_j$; and the state parameters $\boldsymbol{S}^{r+1,n}$. Other than these three groups, the transition probabilities p_{11}, p_{22}, and ν are simulated separately. Detailed of the conditional posterior distributions are as follows:

(1) Let $\boldsymbol{\Theta}_{-\phi_j}$ be the parameter vector $\boldsymbol{\Theta}$ excluding element $\boldsymbol{\phi}_j$. The conditional posterior distribution of $\boldsymbol{\phi}_j$ is given by:

$$p\left(\boldsymbol{\phi}_j \mid \boldsymbol{y}^{r+1,n}, \boldsymbol{S}^{r+1,n}, \boldsymbol{\Theta}_{-\phi_j}\right) \propto p\left(\boldsymbol{y}^{r+1,n} \mid \boldsymbol{S}^{r+1,n}, \boldsymbol{\Theta}\right) p\left(\boldsymbol{\phi}_j\right).$$

(2) The conditional posterior distribution of the variance parameter $\boldsymbol{\alpha}_j$ is

$$p\left(\boldsymbol{\alpha}_j \mid \boldsymbol{y}^{r+1,n}, \boldsymbol{S}^{r+1,n}, \boldsymbol{\Theta}_{-\alpha_j}\right) \propto p\left(\boldsymbol{y}^{r+1,n} \mid \boldsymbol{S}^{r+1,n}, \boldsymbol{\Theta}\right) p\left(\boldsymbol{\alpha}_j\right), \tag{6}$$

which is not of closed form.

(3) The conditional posterior distribution of the transition probability p_{ij} depends only on $\boldsymbol{S}$. Let n_{ii} denote the number of observations in state i at time $t-1$ that remain at state i at time t, and let n_i denote the number observations in state $s_t = i$. The conditional posterior distribution of the probability p_{ii} is $\text{Beta}(u_{i1} + n_{ii}, u_{i2} + n_i - n_{ii})$.

(4) For $t = r+1, \ldots, n-1$, the conditional posterior distribution of state s_t is:

$$\begin{aligned} p\left(s_t \mid \boldsymbol{y}^{r+1,n}, \boldsymbol{S}_{-t}^{r+1,n}, \boldsymbol{\Theta}\right) \quad \propto \quad & \prod_{j=t}^{n} p\left(y_j \mid \boldsymbol{y}^{r+1,j-1}, \boldsymbol{S}^{r+1,j}, \boldsymbol{\Theta}\right) \\ & \times p\left(s_t \mid s_{t-1}, p_{11}, p_{22}\right) p\left(s_{t+1} \mid s_t, p_{11}, p_{22}\right). \end{aligned} \tag{7}$$

When $t = n$, the conditional posterior distribution of state s_n is the same as in (7) but with the last factor $p(s_{t+1} \mid s_t, p_{11}, p_{22})$ omitted. As a result, the conditional posterior probability of $s_t = i$ is:

$$p\left(s_t = i \mid \boldsymbol{y}^{r+1,n}, \boldsymbol{S}_{-t}^{r+1,n}, \boldsymbol{\Theta}\right) = \frac{L(s_t = i) p(s_t = i \mid \boldsymbol{S}_{-t}^{r+1,n}, \boldsymbol{\Theta})}{\sum_{j=1}^{2} L(s_t = j) p(s_t = j \mid \boldsymbol{S}_{-t}^{r+1,n}, \boldsymbol{\Theta})},$$

for $i = 1, 2$ and $t = r+1, \ldots, n$ where

$$L(s_t = i) \propto \exp\left[\sum_{j=t}^{n} -\frac{1}{2}\ln(h_j) - \frac{\nu+1}{2}\ln\left(1 + \frac{(y_j - \mu_j)^2}{(\nu-2)h_j}\right)\right].$$

The state s_t can be drawn from the uniform distribution $\text{U}(0,1)$. We set $s_t = 1$ if a generated random number is less than the probability $p\left(s_t = 1 \mid \boldsymbol{y}^{r+1,n}, \boldsymbol{S}_{-t}^{r+1,n}, \boldsymbol{\Theta}\right)$ and 2 otherwise.

(5) We re-parameterize the degrees of freedom by defining $\delta = \nu^{-1}$. The prior for δ is $\mathrm{U}(0, \frac{1}{4})$ so that $\nu > 4$. This ensures that the first four moments of the distribution of ε_t are finite, and the conditional posterior distribution for δ is:

$$p\left(\delta \mid \boldsymbol{y}^{r+1,n}, \boldsymbol{S}^{r+1,n}, \boldsymbol{\Theta}_{-\delta}\right) \propto p\left(\boldsymbol{y}^{r+1,n} \mid \boldsymbol{S}^{r+1,n}, \boldsymbol{\Theta}\right) p\left(\delta\right).$$

We use the random walk MH algorithm to simulate the parameters from the posterior distributions in (1), (2), and (5) above using a Gaussian proposal to generate the MCMC sample. For the parameters $\boldsymbol{\alpha}$, we choose a combination of the random walk MH algorithm and the independent kernel MH algorithm, as in Chen and So (2006), to help speed up mixing of the MCMC chain.

TESTING MODEL ADEQUACY AND VALUE AT RISK

In recent years there have been many examples of great losses in financial markets, such as the Orange County bankruptcy in the US in 1994 and the collapse of Britain's Barings Bank in 1995. These events caused many financial institutions to put more emphasis on risk management and to avoid improper operations that may create business failure or collapse. Value at risk (VaR) has become a popular approach to measuring the downside risk of a portfolio position in assets. VaR is usually defined as the portfolio loss exceeded with a predetermined probability over a time horizon of k periods:

$$\alpha = \Pr(y_{t+1} + \cdots + y_{t+k} \leq -\mathrm{VaR}).$$

The MCMC method provides a flexible way to determine VaR. Bayesian estimators of the general k-period VaR are constructed via the quantiles of the predictive distribution $p(y_{n+k} \mid \boldsymbol{y})$. In light of this, an α-level VaR estimator can be constructed using the sample α quantiles of the MCMC sample iterates $y_{n+k}^{(j)}$, $j = M+1, \cdots, N$, where M is a burn-in period of MCMC iterations and N is the total number of MCMC iterations, simulated from the forecast density $p(y_{n+k} \mid \boldsymbol{y}, \boldsymbol{\Theta}^{(j)})$. For a one-period VaR under the DMS-GARCH model, we simply simulate:

$$\mathrm{VaR}_{n+1}^{(j)} = -\left[\mu_{n+1}^{(j)} + t_\alpha(\nu^{(j)}) \frac{\sqrt{h_{n+1}^{(j)}}}{\sqrt{\nu^{(j)}/(\nu^{(j)}-2)}}\right],$$

where $\mu_{n+1}^{(j)}$ and $h_{n+1}^{(j)}$ are the conditional mean and variance of $p(y_{n+1} \mid \boldsymbol{y}, \boldsymbol{\Theta}^{(j)})$, $t_\alpha(\nu^{(j)})$ is the αth quantile of a t distribution with $\nu^{(j)}$ degrees of freedom, and $\sqrt{\nu^{(j)}/(\nu^{(j)}-2)}$ is the adjustment term for the standardized t with $\nu^{(j)}$ degrees of freedom. Then we estimate VaR at time $n+1$ as a Monte Carlo average:

$$\mathrm{VaR}_{n+1} = \frac{1}{N-M} \sum_{j=M+1}^{N} \mathrm{VaR}_{n+1}^{(j)}.$$

A good criterion to test VaR forecasts is the number of violations within m testing days. It is defined as the numbers of days in which the loss of actual returns exceeds the forecasted VaR over the specified time horizon. The violation rate is:

$$\hat{\alpha} = \frac{1}{m} \sum_{t=n+1}^{n+m} I\left(y_t < -\mathrm{VaR}_t\right).$$

To obtain the number of violations, we set n observations to be the learning period and m observations to be the testing period. To get $\mathrm{VaR}_t, t = n+1, \ldots, n+m$, we use a moving window forecast. We first obtain the one-step-ahead forecast VaR_{n+1} from $(y_1, \ldots, y_n)$. Next, we obtain VaR_{n+2} from $(y_2, \ldots, y_{n+1})$ and repeat the VaR forecasting m times to get each $\mathrm{VaR}_{n+t}, t = 1, \ldots, m$. To examine the VaR forecasting accuracy, we adopt two back-testing criteria—unconditional and conditional coverage. Under the null hypothesis that $\alpha = \alpha_0$, the expected number of violations is αm. Kupiec (1995) examines whether VaR estimates, on average, provide correct coverage of the lower α tails of the forecasted distributions. Christoffersen (1998) develops a conditional coverage test that examines whether VaR estimates exhibit correct conditional coverage at each point in time. The conditional coverage test examines simultaneously whether the violations appear independently and the unconditional coverage is α. For more properties of these tests see Christoffersen (1998).

SIMULATION STUDY

We perform simulation studies to examine the effectiveness of our MCMC sampling scheme and to see how the two regimes can be differentiated from the suitable posterior distributions. We fix $p = 2$, $g = k = 1$, and the error distribution as t_8^* (i.e., the standardized t distribution with 8 degrees of freedom) in (1). The true values of parameters are reported in Table 1.

To start the MCMC sampling, we set initial values for the parameters in the mean equation to $\boldsymbol{\phi}_1 = \boldsymbol{\phi}_2 = (0, 0, 0)'$ and in the volatility equation to $\boldsymbol{\alpha}_1 = (0.1, 0.1, 0.2)'$ and $\boldsymbol{\alpha}_2 = (0.1, 0.1, 0.1)'$. The initial values of p_{00} and p_{11} are chosen to be 0.5. We randomly select the initial value of the state vector $\boldsymbol{S}^{r+1,n}$ from a first-order Markov process assuming values in $\{0, 1\}$. The choice of prior for the variance intercept $\alpha_0^{(j)}$ is $\mathrm{U}(0, s^2)$, where s^2 is the sample variance. We carry out $N = 30{,}000$ MCMC iterations and use the last 10,000 iterations for posterior inference. We set the sample size at $n = 2{,}000$ and analyze 100 replicated data sets. For sensitivity analysis, we also consider priors of p_{00} and p_{11} to be $\mathrm{Beta}(95, 5)$, $\mathrm{Beta}(19, 1)$, $\mathrm{Beta}(80, 20)$, $\mathrm{Beta}(8, 2)$, and $\mathrm{U}(0, 1)$. For identification purposes, we impose the constraint that $\alpha_1^{(1)} + \beta_1^{(1)} > \alpha_1^{(2)} + \beta_1^{(2)}$.

We construct trace plots and autocorrelograms as diagnostic tools to assess convergence of MCMC chains. Overall, we observe that the proposed sampling scheme generates MCMC iterates that converge quickly to the desired posterior distribution.

Table 1 displays summary statistics of the posterior mean estimator from 100 replications for *Models 1* and *2* using the prior Beta$(8, 2)$ for transition probabilities. The coverage probabilities (CP) are the percentages of 95% posterior intervals, constructed by MCMC iterates, covering the true values. Most biases in the posterior mean estimators of the GARCH parameters are small. One exception is β_1, for which we usually observe a downward bias as in standard maximum likelihood estimation. In general, we do not detect any significant variation among the results generated from the four priors, indicating that the posterior means of the GARCH parameters are fairly robust to the beta priors.

To see how the two regimes can be differentiated from the suitable posterior distributions, summary statistics of the unconditional mean $\mu^{(i)} = \phi_0^{(i)}/(1 - \phi_1^{(i)} - \phi_2^{(i)})$ in *Model 1* and the unconditional variance $\boldsymbol{H}^{(i)} = \alpha_0^{(i)}/(1-\alpha_1^{(i)}-\beta_1^{(i)})$ in *Models 1* and *2* are reported in Table 1 for $i = 1, 2$. It is evident that the two conditional mean structures given by $\phi_1^{(i)}$ and $\phi_2^{(i)}$ are effectively separated since $\mu^{(1)}$ and $\mu^{(2)}$ closely match with their true values. Even though we only use the constraint $\alpha_1^{(1)} + \beta_1^{(1)} > \alpha_1^{(2)} + \beta_1^{(2)}$ in the variance equation for identification, we can identify the two mean regimes through the MCMC sampling. Although $\boldsymbol{H}^{(2)}$ is somewhat overestimated by its posterior mean estimator, it is observed that the two variance regimes are quite well separated by $\boldsymbol{H}^{(1)}$ and $\boldsymbol{H}^{(2)}$, or by $\alpha_1^{(1)} + \beta_1^{(1)}$ and $\alpha_1^{(2)} + \beta_1^{(2)}$. It is anticipated that when the sample size n increases or when the difference between $\boldsymbol{H}^{(1)}$ and $\boldsymbol{H}^{(2)}$ increases, the separation of the two regimes algorithm will be more effective.

EMPIRICAL STUDY

We consider eight daily stock market indices: the CAC40 of France, ADX30 of Germany, Milan MIBTel of Italy, FTSE100 of the UK, Toronto SE300 of Canada, Nikkei225 of Japan, HSI of Hong Kong, and TWSI of Taiwan. Data covering January 4, 1995, to June 29, 2007 were obtained from Datastream International. Daily log returns are calculated as $y_t = (\ln P_t - \ln P_{t-1}) \times 100$, where P_t is the price index at time t.

Descriptive statistics for the daily returns of the eight indexes are given in Table 2. All series have similar means and medians close to zero, minima which vary between -14.73 and -6.04, and maxima that range from 4.68 to 17.25. From the classical mean-variance portfolio assessment, the Canadian market seems to perform the best in the time period considered because Canadian returns have the smallest standard error (SE) and a very large mean. From the historical SEs, the German market returns are the most volatile. This high volatility is consistent with the widest range of returns in the eight markets. We also observe typical negative skewness in all series except the Hong Kong market, so it is not surprising to see that most sample medians are greater than the respective sample means. For kurtosis, all eight return series display the standard property of heavy-tailed distributions. The Jarque-Bera test rejects normality in each market.

Table 1: Summary statistics of the posterior mean estimator for *Models 1* and *2* obtained from 100 replications.

Beta(8, 2)	*Model 1*							*Model 2*					
	True	Mean	SE	2.5%	97.5%	CP		True	Mean	SE	2.5%	97.5%	CP
$\phi_0^{(1)}$	.20	.199	.032	.137	.263	92%	$\phi_0^{(1)}$	.01	.031	.241	−.454	.514	98%
$\phi_1^{(1)}$	−.40	−.378	.046	−.467	−.289	91%	$\phi_1^{(1)}$	−.05	−.035	.143	−.329	.248	100%
$\phi_2^{(1)}$	.20	.213	.038	.139	.287	90%	$\psi_1^{(1)}$	.10	.072	.157	−.236	.405	100%
$\phi_0^{(2)}$	.30	.320	.040	.246	.403	93%	$\phi_0^{(2)}$	.06	.040	.083	−.125	.208	100%
$\phi_1^{(2)}$	.40	.372	.046	.282	.462	89%	$\phi_1^{(2)}$	.02	−.020	.057	−.133	.097	95%
$\phi_2^{(2)}$	.20	.201	.041	.119	.281	96%	$\psi_1^{(2)}$	.30	.261	.076	.130	.432	95%
$\alpha_0^{(1)}$	.07	.107	.036	.053	.193	82%	$\alpha_0^{(1)}$	.15	.311	.198	.049	.808	100%
$\alpha_1^{(1)}$	.05	.098	.034	.039	.174	74%	$\alpha_1^{(1)}$	.10	.141	.066	.033	.297	100%
$\beta_1^{(1)}$	.88	.773	.070	.613	.886	63%	$\beta_1^{(1)}$	.85	.665	.151	.303	.885	93%
$\alpha_0^{(2)}$	.01	.021	.011	.005	.046	89%	$\alpha_0^{(2)}$	.05	.142	.065	.039	.298	78%
$\alpha_1^{(2)}$	.10	.159	.044	.084	.253	79%	$\alpha_1^{(2)}$	.07	.124	.033	.067	.199	58%
$\beta_1^{(2)}$	.80	.744	.053	.628	.833	91%	$\beta_1^{(2)}$	.83	.777	.062	.634	.881	98%
p_{00}	.91	.911	.019	.868	.944	95%	p_{00}	.80	.613	.149	.302	.862	91%
p_{11}	.89	.892	.021	.849	.929	98%	p_{11}	.75	.839	.106	.566	.965	100%
ν	8.00	7.075	1.387	5.068	1.307	83%	ν	8.00	7.621	1.437	5.443	10.988	98%
$\mu^{(1)}$	.17	.171	—	—	—	—							
$\mu^{(2)}$	.75	.749	—	—	—	—							
$\boldsymbol{H}^{(1)}$	1.00	1.085	—	—	—	—	$\boldsymbol{H}^{(1)}$	3.00	2.870	—	—	—	—
$\boldsymbol{H}^{(2)}$	.10	.235	—	—	—	—	$\boldsymbol{H}^{(2)}$	.50	1.944	—	—	—	—

Table 2: Descriptive statistics of daily stock index returns for eight stock markets from January 4, 1995, to June 29, 2007.

	Mean	SE	Skewness	Excess kurtosis	Normality test[a]	Asymmetry test[b]	Min.	Max.
France	.0330	1.3478	−.0314	2.5743	851	139	−6.0448	7.0023
Germany	.0355	1.4890	−.1742	2.9505	1133	218	−8.3822	7.5527
Italy	.0347	1.2144	−.0917	2.7815	993	230	−6.6337	6.8318
U.K.	.0192	1.0577	−.1566	2.8875	1085	186	−5.5890	5.9026
Canada	.0317	.9542	−.6656	6.1064	5013	81	−8.4652	4.6835
Japan	−.0078	1.4056	−.0241	1.8604	430	33	−7.2340	7.6605
Hong Kong	.0255	1.5989	.1377	11.7392	17264	393	−14.7347	17.2471
Taiwan	.0024	1.5556	−.1291	2.6038	847	50	−9.9360	8.5198

[a]The Jarque-Bera normality test statistic. [b]Engle-Ng joint test statistic. All tests are statistically significant with p-values less than 0.0001.

Model fitting

To follow the conventional approach and to provide a basis of comparison, we start with a linear GARCH model as follows:

$$\begin{aligned} y_t &= a_t, \\ a_t &= \sqrt{h_t}\varepsilon_t, \quad \varepsilon_t \sim t_\nu^*, \\ h_t &= \alpha_0 + \alpha_1 a_{t-1}^2 + \beta_1 h_{t-1}. \end{aligned} \tag{8}$$

Bayesian estimates are presented in Table 3, which contains the posterior means together with SEs of α_0, α_1, β_1, and ν. We note the standard empirical result of a high level of volatility persistence in each market, with estimates of $\alpha_1+\beta_1$ between 0.98 and 0.99. To determine whether an asymmetric model is required, asymmetry in volatility is tested using the joint test of Engle and Ng (1993). Table 2 also presents results of the joint test, which indicates significant asymmetric volatility effects in all markets.

Table 3: Posterior mean of the unknown parameters for all markets for the GARCH-t model. SEs are in parentheses.

	France	Germany	Italy	UK	Canada	Japan	Taiwan	Hong Kong
α_0	.013	.018	.020	.008	.008	.037	.044	.008
	(.005)	(.006)	(.006)	(.003)	(.003)	(.014)	(.019)	(.003)
α_1	.065	.088	.108	.077	.082	.064	.074	.076
	(.009)	(.012)	(.015)	(.011)	(.015)	(.012)	(.015)	(.010)
β_1	.930	.906	.884	.917	.911	.921	.914	.917
	(.010)	(.012)	(.016)	(.011)	(.016)	(.015)	(.018)	(.011)
ν	22.359	15.684	10.683	23.892	8.154	8.432	6.375	23.123
	(27.412)	(4.654)	(2.131)	(34.273)	(1.101)	(1.322)	(.880)	(15.633)

We specify the DMS-GARCH model in (1) using Standard and Poor's 500 (S&P500) index return as an exogenous variable:

$$\begin{aligned} y_t &= \begin{cases} \phi_0^{(1)} + \phi_1^{(1)} y_{t-1} + \psi_1^{(1)} x_{t-1} + a_t, & \text{if } s_t = 1, \\ \phi_0^{(2)} + \phi_1^{(2)} y_{t-1} + \psi_1^{(2)} x_{t-1} + a_t, & \text{if } s_t = 2, \end{cases} \\ a_t &= \sqrt{h_t}\varepsilon_t, \quad \varepsilon_t \sim t_\nu^*, \\ h_t &= \begin{cases} \alpha_0^{(1)} + \alpha_1^{(1)} a_{t-1}^2 + \beta_1^{(1)} h_{t-1}, & \text{if } s_t = 1, \\ \alpha_0^{(2)} + \alpha_1^{(2)} a_{t-1}^2 + \beta_1^{(2)} h_{t-1}, & \text{if } s_t = 2. \end{cases} \end{aligned} \tag{9}$$

This model specification includes an autoregressive term and a lagged cross-asset return in the mean equation. The motivation to include one lag of y_t is to check market

efficiency. The S&P500 index return is included to examine any asymmetric spillover effect from the US market. According to the model specification, the dynamic structure of the mean equation is dependent on an autocorrelation term and the "lagged" US market return; the variance equation follows a GARCH(1,1) process for each state. Chen, Chiang and So (2003) employ a double-threshold GARCH model with a US market threshold variable to explore the dynamics of daily stock index returns for six international markets. Their results provide strong evidence supporting an asymmetric nonlinear spillover effect from the US market to other markets in Europe and Asia. Further, Chen and So (2006) explore a range of international markets to use as an exogenous variable and threshold values.

For identification purposes, we again set the restriction $\alpha_1^{(1)} + \beta_1^{(1)} > \alpha_1^{(2)} + \beta_1^{(2)}$ so as to have higher volatility persistence in regime 1. The results (available upon request) suggest that the estimated values of $\beta_1^{(1)}$ and $\beta_1^{(2)}$ are very similar. Therefore, we refit a submodel with the restriction $\beta_1^{(1)} = \beta_1^{(2)}$. This was done by sampling $\alpha_1^{(1)}$, $\beta_1^{(1)}$ and $\alpha_1^{(2)}$ and setting $\beta_1^{(2)} = \beta_1^{(1)}$ in the MCMC procedure. Posterior mean and SEs of fitting (9) are reported in Table 4.

Most of posterior mean estimates of $\phi_1^{(1)}$ and $\phi_1^{(2)}$ are negative while estimates of $\psi_1^{(1)}$ and $\psi_1^{(2)}$ are positive, implying that the previous day's S&P500 return had a positive effect on the direction of the other markets' movement. We find from the variance equations that all persistence parameters $\alpha_1^{(i)} + \beta_1^{(i)}$ are greater than 0.95 except that for Japan. As $\alpha_0^{(1)}$ and $\alpha_0^{(2)}$ are similar, the unconditional variance, characterized by $\boldsymbol{H}^{(i)} = \alpha_0^{(i)}/(1 - \alpha_1^{(i)} - \beta_1^{(i)})$, is larger in regime 1 (i.e., $\boldsymbol{H}^{(1)} > \boldsymbol{H}^{(2)}$). Therefore, it makes sense to regard regime 1 as a higher volatility state. We also observe that both p_{11} and p_{22} are greater than 0.63 with p_{11} generally greater than p_{22}. Therefore, each market had a higher chance of staying in its current volatility state instead of switching regimes.

In terms of the distribution of ε_t, all markets except France and the UK show degrees of freedom ranging from 6 to 14. This suggests the existence of leptokurtosis and, therefore, that it is appropriate to use a heavy-tailed error distribution. For France and the UK, the degrees of freedom are estimated to be around 20, which demonstrates mild excess kurtosis. From the posterior distribution of the parameter estimates in Figure 1, it is clear that the two regimes identified using $\alpha_1^{(1)} + \beta_1^{(1)} > \alpha_1^{(2)} + \beta_1^{(2)}$ show different dynamics in the mean and variance equations. The posterior distributions of $\phi_0^{(2)}$, $\phi_1^{(2)}$ and $\psi_1^{(2)}$ exhibit significant location shift from those of $\phi_0^{(1)}$, $\phi_1^{(1)}$ and $\psi_1^{(1)}$. Therefore, even though the identification mechanism is primarily for volatility persistence, the two regimes also show different mean dynamic structures.

It is important in any econometric study to test for model mis-specification. We perform residual tests of model adequacy. The residuals here are from a standardized t distribution with variance 1; as such they are not Gaussian and the standard Ljung-Box Q statistics cannot be used. We thus turn to the generalized residuals of Smith

(1985). The transformation is:

$$\hat{\varepsilon}_t^{(i)} = \frac{\hat{a}_t^{(i)}}{\sqrt{\hat{h}_t^{(i)}}} \sim t^*_{\nu^{(i)}} \implies \tilde{\varepsilon}_t^{(i)} = \hat{\varepsilon}_t^{(i)} \times \sqrt{\frac{\nu^{(i)}}{\nu^{(i)}-2}} \sim t_{\nu^{(i)}},$$

where $t_{\nu^{(i)}}$ denotes the ordinary t distribution with $\nu^{(i)}$ degrees of freedom, and $\hat{a}_t^{(i)}$ and $\hat{h}_t^{(i)}$ are computed using the ith iterate of $\boldsymbol{\Theta}$. Since $p(t_{\nu^{(i)}} \leq \tilde{\varepsilon}_t^{(i)} \mid \nu^{(i)}) \in \mathrm{U}(0,1)$, we get $\Phi^{-1}[p(\varepsilon_t^{(i)} \leq \tilde{\varepsilon}_t^{(i)} \mid \nu^{(i)})] \sim N(0,1)$. The Ljung-Box tests constructed using the standardized residuals and squared standardized residuals up to the 20th lag are computed. Under model (9), all tests are insignificant at the 1% level. Results are omitted to save space.

Volatility and VaR forecasting

One of the applications of the DMS-GARCH model is to forecast VaR thresholds. Here we compare the performance of five heteroscedastic models. One benchmark model for calculating VaR is RiskMetrics^{TM}, developed by JPMorgan Chase & Co., which is basically the IGARCH model. The RiskMetrics model can be specified as:

$$\begin{aligned} &y_t = a_t, \\ &a_t = \sqrt{h_t}\varepsilon_t, \quad \varepsilon_t \sim N(0,1), \\ &h_t = \lambda h_{t-1} + (1-\lambda)a_{t-1}^2, \quad 0 < \lambda < 1. \end{aligned} \tag{10}$$

The decay factor is set to $\lambda = 0.94$ as suggested by JPMorgan for daily data. The second model we consider is the GARCH-n model defined as:

$$\begin{aligned} &y_t = a_t, \\ &a_t = \sqrt{h_t}\varepsilon_t, \quad \varepsilon_t \sim N(0,1), \\ &h_t = \alpha_0 + \alpha_1 a_{t-1}^2 + \beta_1 h_{t-1}. \end{aligned}$$

We also use the GARCH-t model in (8) for comparison. The fourth model is the IGARCH model with no drift—this is just model (10) with λ unknown rather than fixed at 0.94. The last model considered is the DMS-GARCH model in (9).

We divide our sample into two periods: a learning period from January 4, 1995, to June 23, 2005, and a validation period containing the last 500 observations. The number of trading days in each year is slightly different between markets. A rolling window approach is used to produce one-day-ahead forecasts of h_t and 1% and 5% VaR thresholds using the five models. The rolling window approach means the first n observations are used to estimate the model and to forecast the $(n+1)$th observation, the sample is then rolled forward by one observation so that it ranges from the 2nd to the $(n+1)$th observation and the $(n+2)$th observation is forecasted. This roll-forward process is repeated until the end of the sample. To strike a balance between efficiency

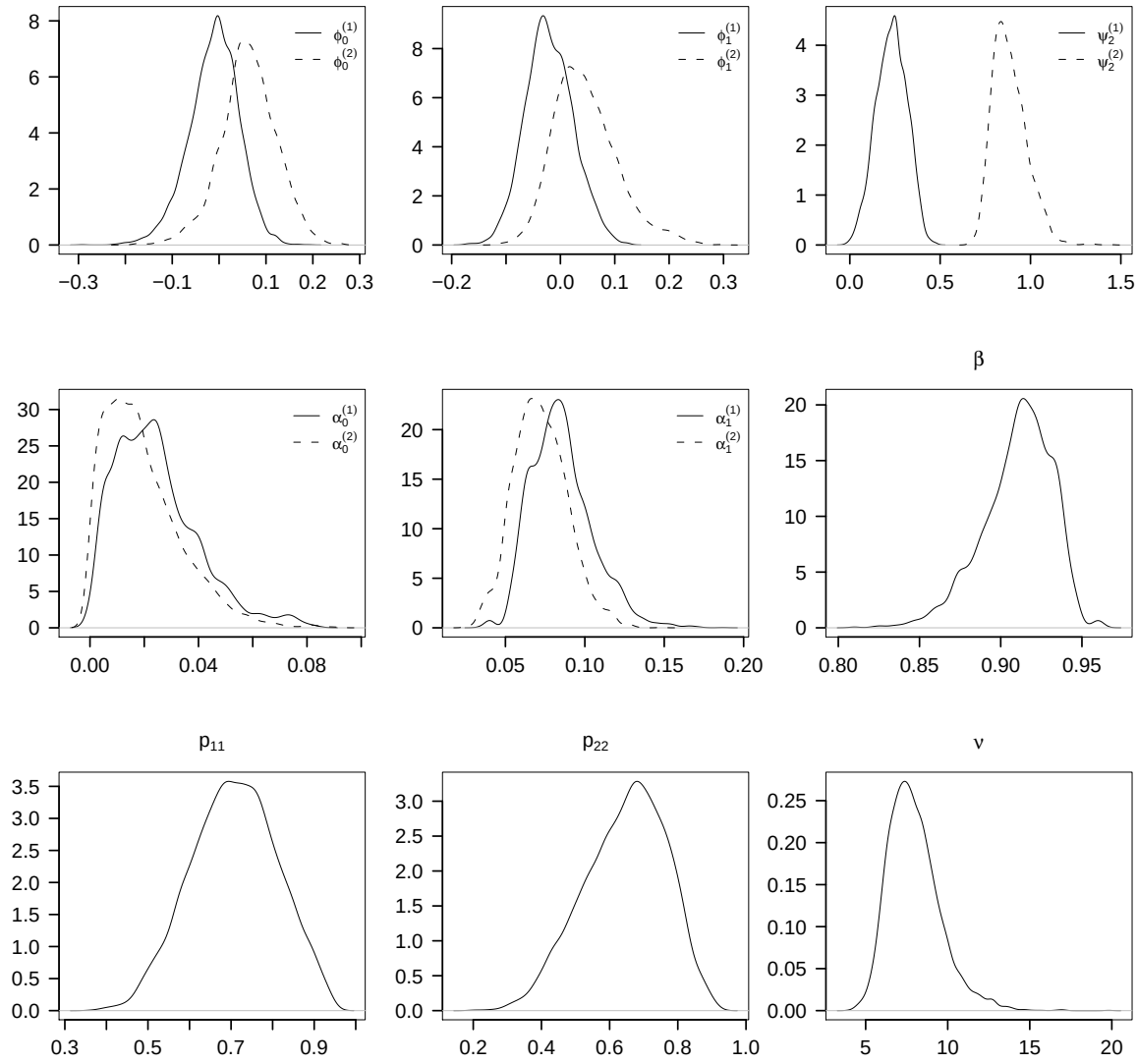


Figure 1: Posterior density of the unknown parameters in the DMS-GARCH model.

Table 4: Posterior mean of the unknown parameters for all markets for DMS-GARCH model. We restrict $\alpha_1^{(1)} + \beta_1^{(1)} > \alpha_1^{(2)} + \beta_1^{(2)}$ and let $\beta_1^{(1)} = \beta_1^{(2)} = \beta$. SEs are in parentheses.

	France	Germany	Italy	UK	Canada	Japan	Taiwan	Hong Kong
$\phi_0^{(1)}$	.063	.081	.035	.045	.064	−.047	.012	−.011
	(.135)	(.126)	(.120)	(.077)	(.074)	(.112)	(.178)	(.053)
$\phi_1^{(1)}$	−.149	−.175	−.077	−.134	.078	−.025	.008	−.020
	(.084)	(.052)	(.054)	(.054)	(.117)	(.074)	(.059)	(.044)
$\psi_1^{(1)}$	.395	.436	.235	.308	.054	.425	.314	.231
	(.143)	(.224)	(.098)	(.070)	(.096)	(.173)	(.103)	(.085)
$\phi_0^{(2)}$	.078	.066	.105	.067	.091	.008	.027	.063
	(.137)	(.167)	(.190)	(.152)	(.102)	(.100)	(.189)	(.060)
$\phi_1^{(2)}$	−.146	−.185	−.066	−.151	−.064	−.086	.009	.046
	(.095)	(.063)	(.080)	(.109)	(.213)	(.061)	(.072)	(.059)
$\psi_1^{(2)}$	.389	.506	.168	.276	.156	.445	.308	.885
	(.167)	(.274)	(.155)	(.140)	(.183)	(.173)	(.145)	(.094)
$\alpha_0^{(1)}$	.027	.017	.033	.019	.017	.312	.086	.024
	(.021)	(.014)	(.025)	(.017)	(.010)	(.302)	(.062)	(.015)
$\alpha_1^{(1)}$	.090	.102	.140	.107	.127	.154	.110	.085
	(.026)	(.027)	(.042)	(.039)	(.034)	(.062)	(.033)	(.019)
β	.903	.896	.854	.887	.867	.782	.874	.910
	(.031)	(.028)	(.045)	(.041)	(.036)	(.119)	(.042)	(.021)
$\alpha_0^{(2)}$	.022	.015	.034	.013	.016	.081	.102	.019
	(.018)	(.012)	(.030)	(.010)	(.011)	(.095)	(.062)	(.014)
$\alpha_1^{(2)}$	.067	.084	.114	.058	.091	.066	.079	.072
	(.028)	(.032)	(.041)	(.037)	(.041)	(.038)	(.028)	(.017)
β	.903	.896	.854	.887	.867	.782	.874	.910
	(.031)	(.028)	(.045)	(.041)	(.036)	(.119)	(.042)	(.021)
p_{11}	.758	.768	.845	.811	.814	.669	.807	.708
	(.162)	(.136)	(.113)	(.140)	(.119)	(.135)	(.117)	(.103)
p_{22}	.741	.734	.779	.642	.718	.635	.748	.645
	(.140)	(.128)	(.118)	(.156)	(.139)	(.206)	(.133)	(.120)
ν	21.611	13.210	9.434	20.306	7.625	8.681	6.052	7.964
	(9.130)	(3.291)	(1.955)	(8.267)	(1.186)	(2.169)	(1.048)	(1.597)
$H^{(1)}$	4.537	8.991	6.714	3.161	2.937	4.900	5.590	7.940
$H^{(2)}$	.945	1.516	1.314	.336	.557	.973	2.317	1.230

in estimation and a feasible number of forecasts, a rolling window size of approximately

2000 observations is chosen, leaving 500 observations to be forecasted.

For illustration, time series plots of the one-day-ahead forecasts of h_t based on the DMS-GARCH and GARCH-t models for Hong Kong are presented in Figure 2. The global trend of the forecasts from the two models are the same, but the DMS-GARCH forecasts have more obvious spikes whereas the GARCH-t forecasts are smoother. Consequently, the DMS-GARCH forecasts exhibit higher volatility of volatility, which may be because the DMS-GARCH allows the volatility dynamic to respond to recent market changes via a state variable.

When the choice of VaR model is unclear, we advise obtaining the violation ratio $\hat{\alpha}/\alpha$ used by Wong and So (2003), where $\hat{\alpha}$ is the empirical coverage representing the proportion of days that the actual return is smaller than the VaR forecast. We expect that a good VaR estimation method will have $\hat{\alpha}/\alpha \approx 1$. Table 5 shows the violation ratios in the eight markets. The "closest-to-one" ratios are highlighted by boxes. It is observed that both DMS-GARCH and GARCH-t perform the best when $\alpha = 1\%$, and RiskMetrics seems to give reliable VaR forecasts for $\alpha = 5\%$. For $\alpha = 1\%$, the performance of DMS-GARCH and GARCH-t are superior in the sense that both sets of forecasts lie in the green zone for each market based on Basel Accord Penalty Zones (see McAleer and Veiga, 2007). The worst model is RiskMetrics, with five markets in the yellow zone.

We rank the five models as in So and Yu (2006); that is, for each market index and given α, the closest-to-one ratio is ranked 1, the next closest-to-one ratio is ranked 2 and so on. The overall rank of the five models for each market in the 500 daily validation period are given in Table 6. Combining the performance for both $\alpha = 1\%$ and $\alpha = 5\%$, RiskMetrics is the worst. With a total rank of 40.5, DMS-GARCH, closely followed by GARCH-t, produces more reliable VaR estimates than other competing models.

We next perform back-testing analysis using the likelihood ratio approach. Table 7 reports p-values of test statistics (including the unconditional coverage test and the conditional coverage test) for the five models. For $\alpha = 1\%$ the RiskMetrics model is clearly rejected in five markets. For a 5% significance level both GARCH models in Taiwan can be rejected. In contrast, we observe good performance in the GARCH-t and DMS-GARCH models and few cases of significant conditional (LR_{cc}) or unconditional (LR_{uc}) likelihood ratio test statistics. The number of significant test statistics (at 5% and 1% levels) for RiskMetrics, GARCH-n, GARCH-t, IGARCH, and DMS-GARCH are 8, 2, 1, 2 and 1, respectively. These results tell us that DMS-GARCH performs the best in terms of the coverage tests with p-values. In short, both GARCH-t and DMS-GARCH models can be recommended for estimating VaR, while the latter also provides some insights into volatility asymmetry.

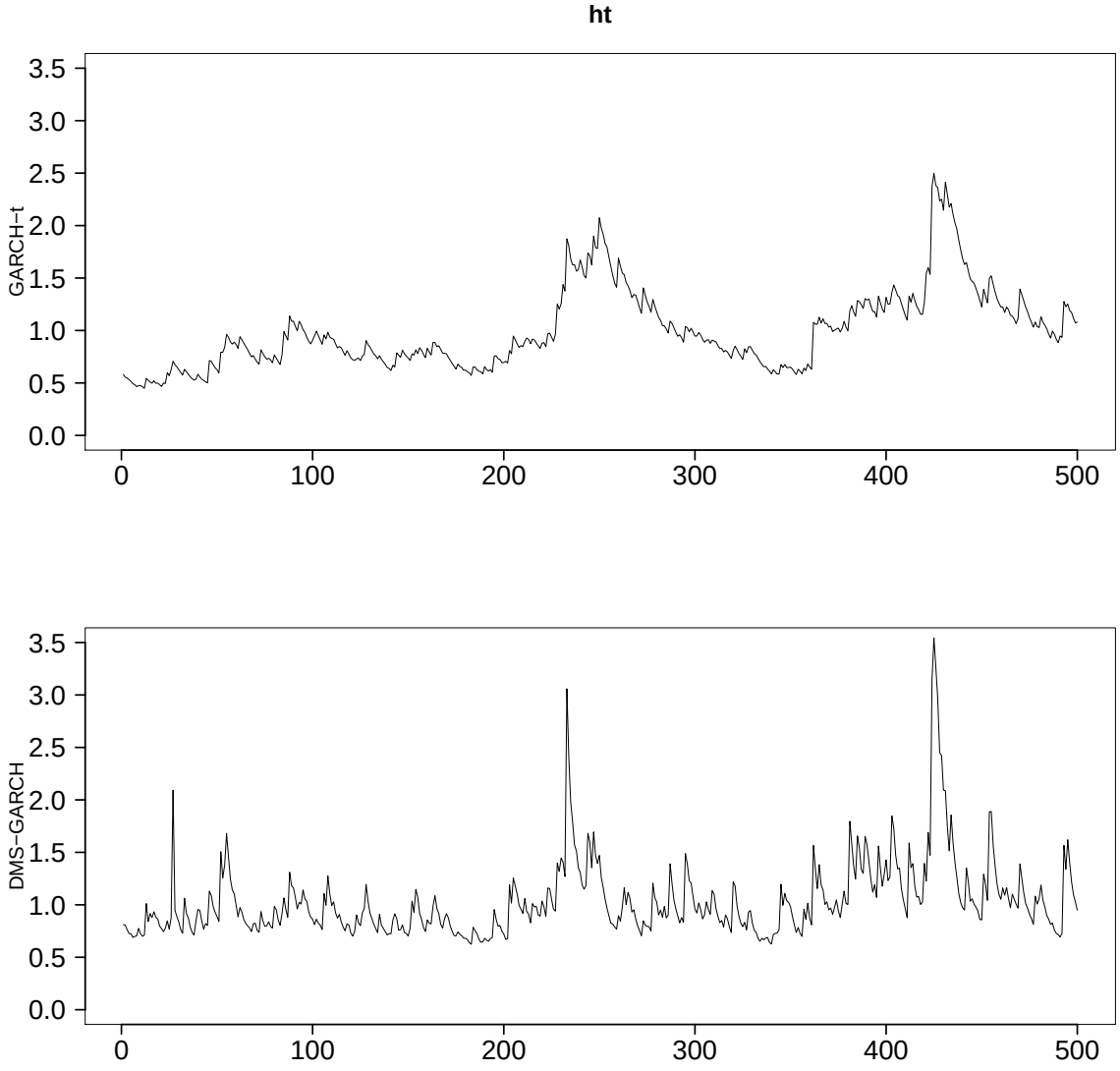


Figure 2: The one-step-ahead volatility forecasts for Hong Kong, derived from $p(\sqrt{h_{n+k}} \mid \boldsymbol{y}^{1,n+k-1})$, for $k = 1, \ldots, 500$, using the GARCH-t and DMS-GARCH models.

Table 5: Ratios of the violation rate over α.

VaR	Market	RiskMetrics	GARCH-n	GARCH-t	IGARCH	DMS-GARCH
α=1%	**France**	1.60	[1.00]	0.80	[1.00]	0.60
	Germany	2.00	1.60	1.60	1.60	[1.00]
	Italy	2.80	[1.40]	[1.40]	[1.40]	1.60
	UK	2.20	1.40	[1.20]	1.40	1.40
	Canada	3.20	2.00	[1.40]	2.00	[1.40]
	Japan	1.60	[1.00]	0.80	1.20	0.80
	Taiwan	2.00	1.60	[1.40]	1.60	[1.40]
	Hong Kong	1.80	1.20	0.80	1.40	[1.00]
α=5%	**France**	[1.08]	0.76	0.80	0.84	0.80
	Germany	[1.04]	0.76	0.80	0.72	1.24
	Italy	1.16	0.76	0.84	0.84	[1.00]
	UK	[1.00]	0.92	0.92	0.92	0.96
	Canada	1.20	[1.12]	1.16	[1.12]	1.36
	Japan	[0.96]	0.80	0.80	0.80	0.88
	Taiwan	[0.92]	0.60	0.60	0.64	0.64
	Hong Kong	1.24	[1.00]	[1.00]	1.08	1.04

CONCLUSION

The proposed double Markov switching GARCH (DMS-GARCH) model is motivated by the advances in the Markov switching model of Hamilton (1989) and the GARCH model of Bollerslev (1986), which assumes that the regime status at time t cannot be observed. The model is designed to capture the main characteristics of asset returns, such as leptokurtosis, volatility clustering, and an asymmetric relationship between stock returns and conditional variance. Estimation and forecasting are performed using Markov chain Monte Carlo methods.

Using eight market indices for illustration, empirical findings show that the fat-tail properties are properly captured by the Markov switching model with t-distributed errors relative to competing models. Our results are based on 500 back-testing days. Kupiec (1995) and Christoffersen (1998) back-testing tests are performed for forecast comparisons of the models. While both the DMS-GARCH and the GARCH-t models are found to be superior to standard approaches, such as RiskMetrics in VaR estimation, the former can also explain possible changes in the volatility dynamic. Extensions to include different state variables in the mean and variance equations and to have more than three regimes are potential directions for further research.

ACKNOWLEDGMENTS

Table 6: The ranks of the models based on VaR violations.

α	Market	RiskMetrics	GARCH-n	GARCH-t	IGARCH	DMS-GARCH
α=1%	**France**	5	1.5	3	1.5	4
	Germany	5	3	3	3	1
	Italy	5	2.5	2.5	2.5	2.5
	UK	5	3	1	3	3
	Canada	5	3.5	1.5	3.5	1.5
	Japan	5	1	3	3	3
	Taiwan	5	3.5	1.5	3.5	1.5
	Hong Kong	5	2.5	2.5	4	1
α=5%	**France**	1	5	3.5	2	3.5
	Germany	1	3.5	2	5	3.5
	Italy	3	5	3	3	1
	UK	5	2.5	2.5	2.5	2.5
	Canada	4	1.5	3	1.5	5
	Japan	1	4	4	4	2
	Taiwan	1	4.5	4.5	2.5	2.5
	Hong Kong	5	1.5	1.5	4	3
	Total	61	48	42	48.5	40.5

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Table 7: P-values of the unconditional and conditional coverage tests.

	RiskMetrics		GARCH-n		GARCH-t		IGARCH		DMS-GARCH	
	LR_{uc}	LR_{cc}	LR_{uc}	LR_{cc}	LR_{uc}	LR_{cc}	LR_{uc}	LR_{cc}	LR_{uc}	LR_{cc}
α=1%										
France	.215	.407	1.000	.951	.641	.869	1.000	.951	.331	.613
Germany	**.048**	.115	.215	.407	.215	.407	.215	.407	1.000	.951
Italy	**.001**	**.003**	.397	.632	.397	.632	.397	.632	.397	.632
UK	**.020**	**.033**	.397	.632	.663	.845	.397	.632	.397	.632
Canada	**.000**	**.000**	**.048**	.115	.397	.632	**.048**	.115	.397	.632
Japan	.215	.407	1.000	.951	.641	.869	.663	.845	.641	.869
Taiwan	**.048**	.115	.215	.407	.397	.632	.215	.407	.397	.632
Hong Kong	.106	.230	.663	.845	.641	.869	.397	.632	1.000	.951
α=5%										
France	.685	.833	.199	.417	.288	.280	.399	.392	.288	.555
Germany	.838	.842	.199	.207	.288	.247	.131	.163	.235	.493
Italy	.423	.424	.199	.207	.399	.279	.399	.279	1.000	.970
UK	1.000	.808	.678	.915	.678	.915	.678	.915	.678	.302
Canada	.319	.189	.546	.177	.423	.188	.546	.177	.079	.118
Japan	.836	.967	.288	.555	.288	.555	.288	.555	.678	.915
Taiwan	.678	.235	**.027**	.066	**.027**	.066	**.049**	.118	**.049**	.084
Hong Kong	.235	.493	1.000	.808	1.000	.808	.685	.833	.838	.234

P-values are from the asymptotic chi-Square distribution of the likelihood ratio test statistic. LR_{uc} denotes the unconditional coverage test statistic proposed by Kupiec (1995). LR_{cc} denotes the conditional coverage test statistic described by Christoffersen (1998).

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Authors' biographies:

Cathy W. S. Chen is a Distinguished Professor of Department of Statistics, Feng Chia University, Taiwan. Her research interests include modeling and forecasting of financial time series, market volatility study, MCMC estimation, and statistical methods in epidemiology. She has published papers in the International Journal of Forecasting, Journal of the Royal of Statistical Society Series C, Computational Statistics & Data Analysis, and Statistics in Medicine, etc.

Mike K. P. So is an Associate Professor of the Department of Information Systems, Business Statistics and Operations Management at the Hong Kong University of Science and Technology. His research interests include nonlinear time series analysis, financial time series modeling, market volatility study, modeling the dynamic structure of economic data and financial risk management.

Edward M. H. Lin is a PhD student in the Graduate Institute of Applied Statistics at Feng Chia University, Taiwan. His research interests include financail time series analysis, forecasting, and Bayesian inference.

Authors' addresses:
Cathy W. S. Chen and Edward M. H. Lin, Graduate Institute of Applied Statistics, Feng Chia University, Taiwan.
Mike K. P. So, the Department of Information Systems, Business Statistics and Operations Management, Hong Kong University of Science and Technology, Clear Water Bay, Hong Kong.